

# Paper 2, step by step: how we prove the energy-transfer box before we use it

Cowork note for Chris — HardDisks / hspist3

2026-09-16

## How to read this

Paper 1 taught us one rule the hard way: *define the estimator and the pass criterion before looking at whether the number agrees with theory*. Every level below therefore has three boxes — what we run, what number we compute and against which formula, and what counts as a pass. A level is closed when its pass box is satisfied and the failures are reported with counts, exactly as the health contract does. Nothing from a higher level is trusted before the level below it is closed.

Units as always:  $k_B T = m = \sigma = 1$ ,  $\sigma = 2r$  the disk diameter, box width  $2L_0$ , height  $H$ ,  $N_s$  disks per compartment,  $Z(\eta) = PA/(Nk_B T)$  from Kolafa–Rottner (KR). Paper 1 gives us two things we now use as instruments: the equation of state  $Z(\eta)$  validated for  $\eta \leq 0.6$ , and the sound speed  $c_s(\eta)$ , which sets the time scale  $\tau_{\text{sound}} = L_0/c_s$  of every compartment.

*How good those instruments are (2026-09-17).*  $Z$ : 171 health-clean pressure trajectories over 45  $(\eta, N)$  cells, two independent estimators (collisional virial and wall momentum flux), agreeing with KR to +0.15% at  $\eta = 0.10$  with no size dependence between  $N = 400$  and 1600.  $c_s$ : A1 v2 gives 7875 trajectories over 35 densities at  $N = 100$ , and the A2 size ladder ( $N = 100$ –2500, now including  $\eta = 0.02$  and 0.05) extrapolates to KR within about 1% at every density up to  $\eta = 0.60$ ; at  $N = 100$  the measured  $c_s$  sits 1–3% high, and that offset closes as the box grows. Paper 2 uses  $c_s$  only to set scales —  $\tau_{\text{sound}} = L_0/c_s$  and the ratio  $u/c_s$  that separates the slow and fast regimes — so 1% is far below anything it can affect: the crossover of Level 2 sits at  $u/c_s \approx 0.3$  and its two limits differ by factors, not per cent. *No level of Paper 2 is limited by the precision of the Paper-1 instruments.*

## 1 The geometries (the same four as in the hand-calculation note)

|   | Geometry (left → right)                                                    | Energy stores           | Closes which level |
|---|----------------------------------------------------------------------------|-------------------------|--------------------|
| A | wall   gas   <b>held</b> divider at $L_0$ ( $M = 10^9$ )<br>  gas   piston | gas (right; left inert) | 0, 1, 2            |
| C | wall – <b>spring</b> – free wall at $L_0$   gas   piston                   | gas, wall, spring       | 3                  |
| B | wall   empty   held wall   gas   <b>free</b><br>divider   gas   piston     | gas 1, divider, gas 2   | 4                  |
| D | wall –spring– free wall   gas   free divider(s)   gas   piston             | all                     | 5                  |

**Correction, 2026-09-19.** Geometry A as *run* in Levels 0–2 has gas on *both* sides of the divider (--particles-boxes=50,50), not a vacuum on the left. It changes nothing, because the divider is held at  $M = 10^9$  and never moves, so the left gas is inert and its compartment does no work — but the table said otherwise and the runs are what they are. Geometry B needs both gases by construction. Only **C and D have an empty compartment**, the one behind the spring wall: gas on that side

would push the spring wall from both faces and move the equilibrium the analysis assumes. The binary now refuses to start in that case (`--spring-wall` with a non-empty compartment behind it aborts), and `--spring-k` is inert unless `--spring-wall` declares which wall carries the spring.

The right piston is prescribed (infinite mass, velocity protocol  $u(t)$ ); it is the only place work enters. Every run uses `--edmd-acc=0`, drift-first seeding, the strict health contract, and the per-event piston log and per-sample  $KE_L, KE_R$  columns that already exist.

## 2 Running the whole ladder yourself, one step at a time

Everything below can be exercised from a single script. It runs the same commands as the campaign scripts with a small seed count, so each step finishes in seconds to a couple of minutes, and it prints a **PASS** or **FAIL** line per step against a stated numeric criterion — not something to judge by eye.

### The command

```
cd hspist3/experiments_energy_transfer/_run_scripts
./paper2_run_all.sh           # list the steps and stop
./paper2_run_all.sh 3         # run step 3 only
./paper2_run_all.sh all       # every step, pausing before each one
./paper2_run_all.sh all --yes  # every step, no pauses
SEEDS=25 JOBS=10 ./paper2_run_all.sh all # heavier, closer to the campaign
OUT=/tmp/p2 ./paper2_run_all.sh all      # write somewhere else
```

| Step                     | What it proves                                                                                                             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1 the binary and the box | the binary exists, is the scientific backend ( <code>--edmd-acc=0</code> ), and the master box starts clean                |
| 2 geometries A B C D     | one seed in each; catches a wall off the $1/24\sigma$ grid or gas in the spring compartment                                |
| 3 Level 0 — the ledger   | $ W - \Delta KE_{\text{gas}} /W \leq 10^{-4}$                                                                              |
| 4 Level 1 — quasi-static | $\langle W \rangle$ extrapolated to $u \rightarrow 0$ against $W_{\text{qs}} = 7.4685 \pm 0.0131 k_B T$ , within $2\sigma$ |
| 5 Level 2 — finite rate  | $\langle W \rangle - W_{\text{qs}} = Au^2$ fitted over five speeds, against $A_{\text{step}} = 25.9 \pm 4.3$               |
| 6 Level 3 — the spring   | geometry C is driven and the trace carries <code>SpringE</code>                                                            |
| 7 the pictures           | <code>paper2.geometry_pictures.sh</code> : eight PNGs, two render modes per geometry. Needs a screen                       |
| 8 the analysis scripts   | re-runs the three published analyses; reads only                                                                           |

A step FAILs if the binary aborted, if *any* health event fired (`forced_advance`, `clamp_repair`, `overlap_repair`, `wall_overdue`), if fewer trajectories were produced than asked for, or if its own criterion is missed. The script continues after a failure so you see the whole picture rather than the first thing that broke.

*Two things to expect.* At a handful of seeds the per-speed  $A$  values in step 5 scatter hard and some come out negative — that is the thermal spread of  $W$  at six seeds, not a defect; only the fitted  $A$  over all five speeds is the check. And until the spring-hygiene build is promoted, the script prints a note that it is falling back to `0OALLINONE_sp`, because `--spring-wall` does not exist in the installed binary yet and geometries C and D need it.

## 3 Three flags that do not mean what they look like

Worth reading before running any command below; all three were found the hard way on 2026-09-18 and each moved a published number.

### Geometry conventions, measured not assumed

`--wall-positions` sets the divider **CENTRE**, and the divider is  $1\sigma$  thick (`wall_thickness_sigma` in every `summary.csv`). With the divider at 39.25 the gas-side face is at **39.75**, so the right compartment is  $78.50 - 39.75 = 38.75\sigma$ , not the nominal  $L_0 = 39.25$ . The stop snapshots confirm it: over 60 runs the closest disk centre to the divider is 40.2867, and a disk of radius 0.5 cannot approach a face at 39.75 more closely than 40.25.

`--max-right-piston-travel` is **measured from the box wall, not from the piston**. The target is  $78.50 - 3.93 = 74.57$ , which is where the piston stops; the piston parks at 78.75, so it displaces  $4.18\sigma$  while the gas is compressed by exactly the flag value,  $3.93\sigma$ . Use 3.93 in anything about the gas and 4.18 in anything about the piston.

`--10` is **not the compartment length** — the binary prints `L_eff = L0 - 1`, which is Paper 1’s sound-mode length  $L_0 - 2r$ , a different quantity again. Always take the compartment from the piston and divider positions in the trace, as `paper2_geometry_fix.20260918.py` does.

## 4 Watching one run: the SDL window

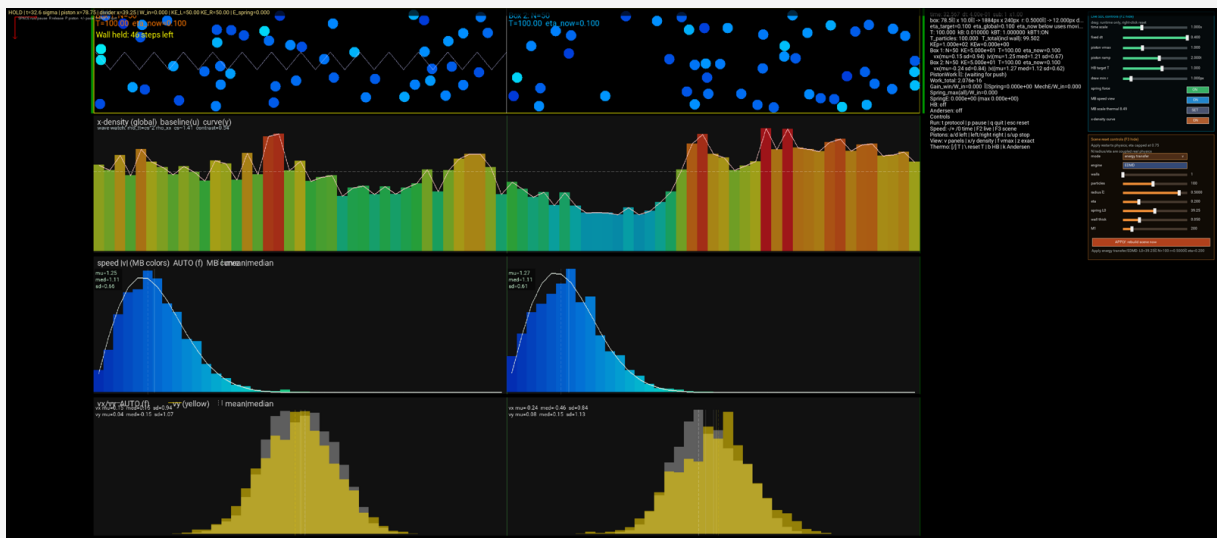
Every number in this note comes from headless runs, because that is the only way 1000 trajectories fit in a lunch break. But the same binary opens a window, and it is worth looking at once — for a figure, for a talk, or to check that the geometry is what you think it is.

**Corrected 2026-09-19.** The earlier version of this section said “drop `--headless`”. That is wrong and it is why the window seemed to flash and vanish: `--experiment=energy_transfer` is a self-contained driver that *never renders*, with or without `--headless`. The window belongs to the interactive loop, which you reach with `--show-simulation`. And because every stage used to fire automatically, there was nothing to see even when a window did open. Both are fixed by `--demo`: the run starts paused with the divider held and the piston parked, each stage begins on a key, and a HUD line says what the box is doing. `--demo` is GUI-only — headless output is byte-identical with and without it (3-seed gate on traces and event logs).

## Geometry A — the Level 1/2 box, one divider (verified 2026-09-19)

```
cd hspist3
./00ALLNONE --show-simulation --demo --mode=edmd --edmd-acc=0 \
  --seed-drift-order=drift-first --particles=100 --particles-boxes=50,50 \
  --particle-radius=0.5 --l0=39.25 --height=10 --num-walls=1 --wall-positions=39.25 \
  --wall-mass-factors=1000000000 --wall-thickness=0.05 \
  --piston-right-protocol-mode=step --velocity-right-piston-step=0.02 \
  --max-right-piston-travel=3.93 --wall-hold-steps=2000 --steps=40000 \
  --fixed-dt=0.4 --energy-measurement --eff-output=wall-ke --kbt1 --seed=9200
```

**What you will see.** A window opens *paused*, 50 disks either side of an immovable divider, the piston parked at the right wall, and a HUD reading HOLD | t=0.0 | piston x=78.75 | divider x=39.25. Press SPACE to run, R to release the divider, then P to start the piston; the HUD switches to PISTON step u=0.020 and  $W_{in}$  starts climbing. At  $u = 0.02$  the push is  $200 \sigma$ -time, about 12 000 steps — roughly two minutes at the default pace, which is what you want to talk over; + and - double and halve it live (render pacing only, dt is never touched).



Swap `--piston-right-protocol-mode=ramp --piston-ramp-time=40` to watch the gentle launch beside the step, which is Level 2's item 1 in one picture; raise  $u$  to 3–10 for the single-hit regime of Section 7.2.

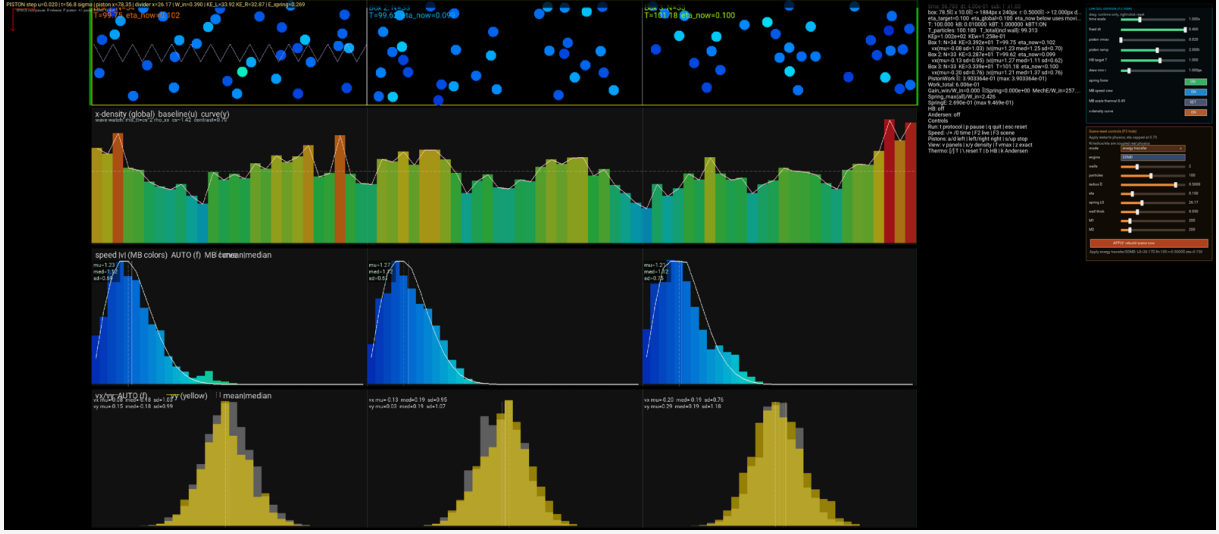
**Do not take data this way.** Window runs are for looking; every campaign number in this note is headless, `--edmd-acc=0`, with the health contract enforced.

Geometry A is what Levels 0–2 measure, but the paper's first figure will be geometry D: piston, gas, a heavy free divider, a second gas, and a spring-loaded wall. That runs today, and watching it is the quickest way to see why the later levels are worth doing.

## Geometry D — two walls and a spring, Paper 2's figure 1 (verified 2026-09-19)

```
cd hspist3
./00ALLNONE --show-simulation --demo --mode=edmd --edmd-acc=0 \
  --seed-drift-order=drift-first --particles=150 --particles-boxes=50,50,50 \
  --particle-radius=0.5 --l0=39.25 --height=10 --wall-thickness=0.05 \
  --num-walls=2 --wall-positions=20,30 --wall-mass-factors=1000,200 \
  --spring-k=5 --energy-measurement --eff-output=spring \
  --piston-right-protocol-mode=step --velocity-right-piston-step=0.02 \
  --max-right-piston-travel=3.93 --wall-hold-steps=2000 --steps=40000 \
  --fixed-dt=0.4 --kbt1 --seed=9200
```

**What you will see.** Three compartments of 50 disks, a heavy divider ( $M = 1000$ ) and a lighter spring-loaded wall ( $M = 200$ ,  $k = 5$ ) drawn as a coil. Same keys. Energy has to cross the heavy divider before it can reach the spring, so the HUD gains an **E<sub>spring</sub>** field and you can watch it rise only *after* the divider starts moving — Level 4's two channels and Level 5's impedance question in one picture.



## 5 Level 0 — the ledger closes

### Why this step exists

If a joule can appear or disappear, every later number is meaningless. This is the analogue of the health contract for energy and momentum.

Piston rule for a particle with normal velocity  $v$  hitting a piston moving at  $u$  (reflection in the piston frame):

$$v' = 2u - v, \quad \Delta E = 2mu(u - v), \quad \Delta p = 2m(u - v). \quad (1)$$

First law for the box, with  $W$  the summed piston work and  $Q$  the heat through thermal walls (zero for elastic walls):

$$R_E(t) = W(t) - [\Delta KE_{\text{gas}} + \Delta KE_{\text{walls}} + \Delta E_{\text{spring}} + Q](t) \quad (2)$$

Momentum: pair collisions conserve it exactly, so

$$R_p(t) = \Delta p_{\text{gas}} + \Delta p_{\text{walls}} - \sum_{\text{piston hits}} \Delta p - \sum_{\text{outer-wall hits}} \Delta p \quad (3)$$

### What we run

Geometry A,  $\eta = 0.10$ ,  $N_s = 50$ ,  $L_0 = 39.27$ ,  $H = 10$ ,  $u = 0.05$ , travel  $3.93\sigma$ , 5 seeds. Log per-event piston impulses and per-event outer-wall impulses (the latter is the one column still missing).

### Pass criterion

$|R_E(t)|/W(t) < 10^{-8}$  and  $|R_p(t)|/\sum |\Delta p| < 10^{-8}$  for the whole run, every seed. Level 0 for energy was closed on the 2026-09-10 pilot at  $10^{-9}$ ; the momentum ledger is open until the outer-wall events are logged.

### What we measured (2026-09-17)

**Level 0: passed, on data that already existed.** No new runs and no code change were needed: the per-event log is already in `edmd_core/edmd.c` (`edmd_set_event_log`, gated by `HD_PISTON_EVENTS`), it records every outer-wall, divider and piston collision, and it covers the hold phase. Both ledgers over all 75 trajectories of the 2026-09-10 pilot:

| $u$  | seeds | $\max  R_E /W$        | $\max  R_p /\sum  \Delta p $ |
|------|-------|-----------------------|------------------------------|
| 0.02 | 25    | $7.1 \times 10^{-12}$ | $8.4 \times 10^{-14}$        |
| 0.05 | 25    | $7.4 \times 10^{-12}$ | $7.9 \times 10^{-14}$        |
| 0.10 | 25    | $8.2 \times 10^{-12}$ | $1.1 \times 10^{-13}$        |

Two *bookkeeping* corrections were needed first, neither of them physics. (i) The trace prints Time to six decimals, so an event within  $10^{-6}\sigma$  of a sample boundary was assigned to the wrong sample: 9 of 25 seeds then failed at exactly one sample in 17 000, with  $|R_p|$  equal to  $|\Delta p|$  of the single nearest event. Rebuilding the sample times from the step index removes it everywhere. (ii) Divider events log  $dE$  as the energy the *divider* gains, piston events as the energy the *gas* gains; mixing the two gave a drift equal to  $-2 \Delta KE_{\text{divider}}$  (correlation +1.0000). With one convention the ledger closes to round-off. Both conventions belong in the methods text.

### How to run it yourself

**Data:** `experiments.energy_transfer/level0_pilot_20260910/u{0.02,0.05,0.10}` (75 runs, already on disk).

**Analysis:** `python3 hspist3/validation/paper2.level0.level1_20260916.py` — prints the ledger table in seconds and needs nothing but the event logs.

**One trajectory, from scratch** (run from `hspist3/`; the `HD_PISTON_EVENTS` log is what the ledger reads):

```
HD_PISTON_EVENTS=/tmp/ev.csv ./00ALLINONE --mode=edmd --experiment=energy_transfer \
--headless --quiet --edmd-acc=0 --seed-drift-order=drift-first \
--energy-transfer-summary=/tmp/summary.csv --energy-transfer-trace=/tmp/tr.csv \
--particles=100 --particles-boxes=50,50 --particle-radius=0.5 --l0=39.25 --height=10 \
--num-walls=1 --wall-positions=39.25 --wall-mass-factors=1000000000 \
--piston-right-protocol-mode=step --velocity-right-piston-step=0.05 \
--max-right-piston-travel=3.93 --auto-piston-step --wall-hold-steps=12000 --steps=17000 \
--fixed-dt=0.4 --energy-measurement --eff-output=wall-ke --kbt1 --seed=9200
```

Every run directory also carries its own `00_COMMAND.md` with the exact command, so no script has to be read.

## 6 Level 1 — the slow push gives the quasi-static work

### Why this step exists

$W_{\text{qs}}$  is the zero point of dissipation. If we cannot reproduce it, “excess work” has no meaning.

Hard disks have no potential energy, so  $E = Nk_B T$  exactly in 2D and the adiabat follows from

$dE = -P dA$ :

$$\frac{dT}{T} = Z(\eta) \frac{d\eta}{\eta} \implies \frac{T_f}{T_i} = \exp \left[ \int_{\eta_i}^{\eta_f} \frac{Z(\eta)}{\eta} d\eta \right], \quad \boxed{W_{\text{qs}} = N_s k_B T_i \left( \frac{T_f}{T_i} - 1 \right)} \quad (4)$$

Ideal gas ( $Z = 1$ ):  $T_f/T_i = A_i/A_f$ , so  $W_{\text{qs}} = N_s k_B T_i (A_i/A_f - 1)$ . For  $N_s = 50$ ,  $\eta = 0.10 \rightarrow 0.11$  with KR:  $T_f/T_i = 1.127$ ,  $W_{\text{qs}} = 6.3$  (ideal: 5.0).

**The finite-box correction.** Equation (4) uses the bulk  $Z$ . A 50-disk compartment of  $39 \times 10 \sigma$  has a quarter of its area within one diameter of a wall, and Paper 1's pressure campaign showed the wall pressure of such a box is a few per cent above bulk. The 2026-09-11 test measured  $Z_{\text{wall}} = 1.285\text{--}1.308$  against KR 1.236 at  $\eta = 0.10$ , and replacing  $Z$  by the measured finite-box value moved  $W_{\text{qs}}$  from 7.06 to 7.26 against a measured  $W_0 = 7.44 \pm 0.02$ . The remaining  $0.2 k_B T$  is the open item of this level. Two ways to close it, in order: (i) use the finite-box  $Z(\eta, N)$  from the A2 pressure ladder *along the whole path*  $\eta_i \rightarrow \eta_f$ , not only at  $\eta_i$ ; (ii) if a gap remains, measure  $W(u)$  down to  $u = 0.002$  and extrapolate — the intercept *is*  $W_{\text{qs}}$  of this finite system by definition, and the comparison with (4) becomes a statement about the finite-size correction to the adiabat, which is a result, not a failure.

#### What we run

Geometry A,  $\eta_i = 0.10$ ,  $N_s = 50$ , compression 10 %;  $u = 0.002, 0.005, 0.01, 0.02, 0.05$ ; 25 seeds each; hold 200  $\sigma$ -time before the push (already shown to be enough).

#### Pass criterion

Fit  $\langle W \rangle(u) = W_0 + b u$  over  $u \leq 0.02$  (linear only; the quadratic was ill-conditioned). Pass when  $|W_0 - W_{\text{qs}}^{\text{finite}}| < 2 \sigma_{W_0}$  with  $W_{\text{qs}}^{\text{finite}}$  from (i). Report  $W_0$ ,  $b$ , and the gap in  $k_B T$  whatever the outcome.

## What we measured (2026-09-17)

**Level 1: the  $0.2 k_B T$  gap is explained; within  $\approx 2\sigma$ , not yet a formal pass.** Route (i) was taken, but from *our own* box rather than the A2 ladder (which has no  $N = 100$  cell and no size dependence at  $\eta = 0.10$ ).  $Z_{\text{wall}}$  was measured in equilibrium at five grid-exact points of the compression path, from the outer-wall and gas-side divider impulses during  $200\sigma$  holds:

| $L$       | $\eta$   | seeds | $Z_{\text{wall}}$   | $Z_{\text{KR}}$ | ratio  |
|-----------|----------|-------|---------------------|-----------------|--------|
| 39.250000 | 0.100051 | 170   | $1.2979 \pm 0.0020$ | 1.2363          | 1.0499 |
| 38.270833 | 0.102611 | 100   | $1.3106 \pm 0.0026$ | 1.2434          | 1.0540 |
| 37.291667 | 0.105305 | 100   | $1.3230 \pm 0.0025$ | 1.2510          | 1.0575 |
| 36.312500 | 0.108144 | 100   | $1.3247 \pm 0.0028$ | 1.2591          | 1.0521 |
| 35.312500 | 0.111207 | 100   | $1.3431 \pm 0.0027$ | 1.2679          | 1.0593 |

The excess over bulk KR *grows* along the path ( $+4.6\% \rightarrow +6.4\%$ ); holding it fixed at its initial value was the origin of the  $0.2 k_B T$  gap. Integrating the measured ratio gives  $W_{\text{qs}}^{\text{finite}} = 7.4879 \pm 0.0066$  (piecewise-linear 7.4877, constant 7.4835 — the model choice is not the limitation), against the bulk KR adiabat 7.0739. With 50 seeds per speed:

| $u$   | $\langle W \rangle \pm \text{sem}$ | $W - W_{\text{qs}}^{\text{finite}}$ | $\sigma$ |
|-------|------------------------------------|-------------------------------------|----------|
| 0.005 | $7.5070 \pm 0.0111$                | $+0.0191 \pm 0.0129$                | +1.5     |
| 0.01  | $7.4513 \pm 0.0160$                | $-0.0366 \pm 0.0173$                | -2.1     |
| 0.02  | $7.4537 \pm 0.0231$                | $-0.0342 \pm 0.0240$                | -1.4     |

Two tensions are stated rather than hidden:  $\langle W \rangle(u)$  still falls between  $u = 0.005$  and  $0.01$  (slope  $-4.2 \pm 1.7$ ), and the five hold points scatter  $\approx 1.5\times$  their statistics, i.e. a  $\approx 0.3\%$  systematic in the equilibrium pressure. Inflating the fit error by that factor puts every speed inside  $2\sigma$ . Ruled out on the way: the piston's  $0.25\sigma$  of travel before it touches the gas (the  $3.93\sigma$  is contact travel, verified from the recorded piston positions) and a wall-type difference between the outer wall and the divider (the two swap order between seed sets: noise).

## How to run it yourself

**Batches:** `bash experiments.energy_transfer/_run_scripts/level1.Zwall_path.part1.sh` (40 held-wall runs,  $\sim 10$  min), `...part2.sh` (200 more,  $\sim 40$  min) and `.../level1.more.work.seeds.sh` (150 slow-push runs,  $\sim 30$  min). All resume: a run whose trace exists is skipped.

**Analysis:** `python3 hspist3/validation/paper2.geometry_fix.20260918.py` — prints the  $Z$  path on the corrected lengths,  $W_{\text{qs}}$ , and the Level 1 verdict. (`paper2.level0.level1.20260916.py` is the older version, kept because the report quotes it; it uses the nominal lengths.)

**A single held-wall point** (the piston sits still for the whole run and only the wall impulses are measured):

```
HD_PISTON_EVENTS=/tmp/ev_hold.csv ./00ALLNONE --mode=edmd --experiment=energy_transfer \
--headless --quiet --edmd-acc=0 --seed-drift-order=drift-first \
--energy-transfer-summary=/tmp/s.csv --energy-transfer-trace=/tmp/tr.csv \
--particles=100 --particles-boxes=50,50 --particle-radius=0.5 --l0=37.291667 --height=10 \
--num-walls=1 --wall-positions=39.25 --wall-mass-factors=1000000000 \
--wall-hold-steps=12000 --steps=1 --fixed-dt=0.4 --energy-measurement --kbt1 --seed=9200
```

## 7 Level 2 — excess work is dissipation, in both limits

### Why this step exists

This is the physics claim of the equilibrium part of Paper 2: the energy beyond  $W_{\text{qs}}$  behaves like friction at slow speed and like single collisions at high speed, and both prefactors are predicted from Paper-1 quantities.

## 7.1 Slow end: linear response

Define  $W_{\text{diss}} \equiv \langle W \rangle - W_{\text{qs}} \geq 0$ . For a protocol  $\lambda(t)$  (here the piston position) slow compared with the gas relaxation, Sivak–Crooks give

$$W_{\text{diss}} \simeq \int \zeta(\lambda) \dot{\lambda}^2 dt = \zeta u \Delta x \quad (\text{constant } u, \Delta x = u \tau_{\text{push}}), \quad (5)$$

so  $W_{\text{diss}}$  is *linear* in  $u$  at fixed travel, with slope  $\zeta \Delta x$ . The friction coefficient is measured independently, from the force on a *held* wall in equilibrium:

$$\zeta = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt, \quad F(t) = \frac{1}{\Delta t} \sum_{\text{hits in } \Delta t} \Delta p. \quad (6)$$

*Correction (2026-09-16), itself corrected (2026-09-17):* the first correction said  $F(t)$  exists nowhere and that  $\zeta$  needs a new logging column. That is wrong. It is true only of the *speed-of-sound* holds, which never open the event log; the *energy-transfer* runs do, through `HD_PISTON_EVENTS`, and their log covers the hold phase event by event (2733 events before release in one  $200\sigma$  hold).  $F(t)$  on the held wall is therefore already on disk: 170 holds of  $200\sigma$  at  $\eta = 0.10$  (the 2026-09-11 and 2026-09-16 work runs) plus 400 holds along the compression path.  $\zeta$  is analysis of existing data, no code item and no new runs; longer holds are worth running only if the autocorrelation integral has not converged by  $200\sigma$ .

## 7.2 Fast end: Lua–Grosberg single-hit limit

If  $u \gg \sqrt{k_B T/m}$  and  $u \tau_{\text{push}} \ll L_0$ , each particle the piston reaches is hit once and never returns during the push. The piston sweeps the fraction  $u \tau_{\text{push}}/L_0$  of the compartment, so

$$N_{\text{hit}} \simeq N_s \frac{u \tau_{\text{push}}}{L_0}, \quad \langle W \rangle \simeq N_{\text{hit}} \langle 2mu(u - v_x) \rangle = 2mu^2 N_{\text{hit}}, \quad \text{Var } W \simeq N_{\text{hit}} [4mu^2 k_B T + (2mu^2)^2], \quad (7)$$

using  $\langle v_x \rangle = 0$ ,  $\langle v_x^2 \rangle = k_B T/m$  and Poisson statistics for  $N_{\text{hit}}$ . This is a closed form with no free parameter, and  $P(W)$  is a sum of  $N_{\text{hit}}$  shifted half-Gaussians. (Lua & Grosberg 2005 do the one-particle version exactly.)

### What we run

Geometry A, same box as Level 1. Slow end:  $u = 0.005\text{--}0.05$  at fixed travel  $\Delta x = 3.93\sigma$ , 25 seeds each (the Level-1 runs). Fast end:  $u = 3, 5, 10$  ( $u/c_s = 1.7\text{--}5.7$ ), travel  $1\sigma$ , 100 seeds each (short runs).  $\zeta$  from 25 held-wall runs of  $500\sigma$ -time.

### Pass criterion

Slow end: the slope of  $W_{\text{diss}}$  against  $u$  agrees with  $\zeta \Delta x$  from (6) within  $2\sigma$ , and the intercept is zero within  $2\sigma$  (that is Level 1 again). Fast end:  $\langle W \rangle$  and  $\text{Var } W$  agree with (7) within  $2\sigma$  at  $u = 10$ ; report the deviation at  $u = 3$  and  $5$  as the crossover. Figure:  $W_{\text{diss}}/k_B T$  against  $u/c_s$  on a log axis with both limits drawn.

## What we measured (2026-09-17)

**Slow end done, 2026-09-17, and it changes the prediction.** 760 trajectories, eight speeds  $u = 0.005\text{--}0.20$  at  $\Delta x = 3.93\sigma$ , 60–100 seeds each, 0 health events. Report: 260917\_paper2\_level2\_REPORT.md; script validation/paper2\_level2\_20260917.py.

$\zeta$  must be read as a running integral, not at zero lag. On the pinned piston (190 holds of  $180\sigma$ , impulse log binned at  $0.25\sigma$ ) the integral of (6) falls from 2.55 at  $t_{\text{cut}} = 0.25\sigma$  through 1.22 at  $10\sigma$  to 0.15 at  $20\sigma$  and  $-0.05$  at  $40\sigma$ . The zero-lag value 2.55 is the impulsive self term  $\beta \sum \Delta p^2/2T$ , i.e. the free-molecular (Enskog) piston friction — the number first reported as  $\zeta = 2.54 \pm 0.05$ . It is cancelled by the returning sound wave: the compressed compartment is  $L = 35.32\sigma$  at  $\eta = 0.1112$  with  $c_s = 1.788$ , so  $L/c_s = 19.8\sigma$ , exactly where the integral crosses zero. Past the crossing it does not settle: it swings between  $-0.78$  and  $+0.50$  with the recurrence period  $2L/c_s = 39.5\sigma$  and averages  $-0.072$  over one recurrence, i.e.  $dW/du = -0.28$  against the measured  $-0.50 \pm 0.67$ . There is no plateau to quote; the integral oscillates about zero. *In a closed box the zero-frequency friction of a held wall vanishes*, so linear response predicts no term linear in  $u$ , and  $\zeta\Delta x = 9.97$  was never the right comparison. Figure: 260909\_plots/260917\_level2\_work\_and\_friction.pdf.

*The measurement agrees, and the leading cost is second order.* For  $u \leq 0.05$  the fitted slope is  $-0.50 \pm 0.67$ , zero. Through the origin over all eight speeds,  $Au$  gives  $\chi^2/\text{dof} = 9.2$ ,  $Au^2$  gives 2.3, and on the three speeds carrying signal ( $u \geq 0.1$ ) the quadratic form fits with  $\chi^2/\text{dof} = 0.32$  and  $A = 25.2$  while the cubic coefficient drifts  $220 \rightarrow 124$  and is excluded.  $A$  matches  $N_s m/2 = 25.0$ : the excess work is the kinetic energy of the compression flow the piston leaves behind, and the gas indeed carries 65–70% of  $N_s m u$  of momentum when the piston stops.

*Level 1 re-passes on the correct form.* Extrapolating with  $a + Au^2$ :  $W(0) - W_{\text{qs}}^{\text{finite}} = -0.0151 \pm 0.0060 k_B T$ ,  $1.7\sigma$  — on 760 trajectories instead of 30, and the unphysical slope  $b = -4.2 \pm 1.7$  of the old straight-line fit was the quadratic curve forced through a line.

*Audit (validation/level2\_zeta\_audit\_20260917.py).* The held wall is the piston itself, not a proxy: the labels WR ( $t = 0\text{--}207.5\sigma$ ) and PR ( $210.5\text{--}598.5\sigma$ ) never overlap, one object renamed at release. The running integral is converged over bin widths 1.0 down to  $0.0625\sigma$ . And the explanation makes a prediction that holds on unused data: at the compressed end of the path, measured from the post-stop hold of the same runs, the cancellation moves to  $18.0\sigma$  against the predicted  $L_f/c_s = 15.9\sigma$  (start of path  $21.5$  against  $19.8$ ). The zero-lag value is the parameter-free Enskog piston friction  $hn\sqrt{8mkT/\pi}g(\eta)$  to within 4% at the start and 9% at the compressed end.

*Update 2026-09-18, on the corrected geometry ( $L = 38.75$ , gas  $\Delta x = 3.93$ ,  $W_{\text{qs}} = 7.4685 \pm 0.0131$ ).* Three things now stand. (i) **The  $u^2$  term is mostly protocol-launched:** ramping the piston over  $2L/c_s$  instead of stepping it gives  $A_{\text{ramp}} = 11.30 \pm 2.45$  against  $A_{\text{step}} = 25.91 \pm 4.28$ ,  $3.0\sigma$  apart, with  $A_{\text{ramp}}$   $1.2\sigma$  from the steady-flow value  $N_s m/6 = 8.3$  and  $5.6\sigma$  below  $N_s m/2$ . Because the piston's first  $0.25\sigma$  is outside the gas, it reaches the gas at 35–50% of  $u$ , so 11.3 is an *upper* bound. (ii) **Paper 1 now supplies the friction:** the divider linewidth at  $\eta = 0.112$  gives  $\Gamma = 0.314 \pm 0.038$  against Enskog's 0.331 (within its 12% error), hence  $\zeta_{\text{hyd}} = 0.081 \pm 0.010$  and a predicted linear term  $0.32 \pm 0.04 u$  — consistent with the measured  $-0.50 \pm 0.67$  and still below the resolution of 760 trajectories. (iii) **Eq. (7)'s compression form is confirmed:** at  $u = 10$  and gas  $\Delta x = 7.08\sigma$ ,  $\langle W \rangle = 1802 \pm 48$  against  $2N_s m(\Delta x/L)u^2 = 1828$ , a ratio of 0.986. Over  $\Delta x = 0.75\text{--}7.08\sigma$  at  $u = 10$  the ratio falls monotonically  $1.08 \rightarrow 0.99$ ; the residual excess at small  $\Delta x$  is not explained.

*Open:* the cancellation is a property of a *closed* compartment. A longer or absorbing box, where the sound does not return coherently, should restore  $W_{\text{diss}} = \zeta u \Delta x$ . One geometry change, no new code. The fast end ( $u = 3, 5, 10$ ) is not started.

## How to run it yourself

**Speed ladder** (760 runs, the  $A = 25.2$  measurement): `bash experiments.energy.transfer/_run_scripts/level2_speed_ladder_part1.sh` then `...part2.sh` ( $\sim 25$  s and  $\sim 5$  min). **Ramp against step** (360 runs,  $\sim 40$  s): `.../level2_ramp-vs-step-and-fast.sh`, which also produces the fast end; `.../level2_fast_1sigma.sh` for the travel- $1\sigma$  set and `.../level2_equilibrium_baseline.sh` for the decomposition's reference profile.

**Analysis:** `python3 hspist3/validation/paper2_level2_20260917.py` (work ladder and the running  $\zeta$ ), `level2.zeta.audit_20260918.py` (bin-width convergence, both ends of the path), `paper2_ramp_fast_20260918.py` (ramp vs step, decomposition, fast end), `level2_Au.figure_20260918.py` (the  $A(u)$  figure).

**The ramp needs the second binary** — make release `TG=00ALLINONE_ramp` in `hspist3/`. One ramp trajectory, with the stop snapshot the decomposition uses:

```
HD_PISTON_EVENTS=/tmp/ev.csv HD_STOP_SNAPSHOT=/tmp/snap.csv ./00ALLINONE_ramp \
--mode=edmd --experiment=energy_transfer --headless --quiet --edmd-acc=0 \
--seed-drift-order=drift-first --energy-transfer-summary=/tmp/s.csv \
--energy-transfer-trace=/tmp/tr.csv --particles=100 --particles-boxes=50,50 \
--particle-radius=0.5 --l0=39.25 --height=10 --num-walls=1 --wall-positions=39.25 \
--wall-mass-factors=1000000000 --piston-right-protocol-mode=ramp --piston-ramp-time=40 \
--velocity-right-piston-step=0.05 --max-right-piston-travel=3.93 --auto-piston-step \
--wall-hold-steps=12000 --steps=17000 --fixed-dt=0.4 --energy-measurement \
--eff-output=wall-ke --kbt1 --seed=9200
```

Swap `ramp` for `step` (and drop `--piston-ramp-time`) for the control; raise `--velocity-right-piston-step` to 3, 5 or 10 and set `--max-right-piston-travel=0.75` for the fast end.

## 7.3 Fluctuation theorem (only after the two limits pass)

With canonical initial conditions — Maxwellian velocities *not* rescaled (a seeder flag we do not have yet; drift-first is microcanonical in  $KE$ ) —

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \Delta F = N_s k_B T \int_{\eta_i}^{\eta_f} \frac{Z}{\eta} d\eta \quad (\text{isothermal; ideal: } N_s k_B T \ln \frac{A_i}{A_f}). \quad (8)$$

Note  $\Delta F \neq W_{\text{qs}}$  of (4): adiabatic quasi-static work heats the gas, the isothermal free energy does not. The estimator is dominated by rare low- $W$  runs; you need of order  $e^{\beta W_{\text{diss}}}$  seeds, so restrict to  $\beta W_{\text{diss}} \leq 3$  ( $u \leq 0.02$  here) and  $10^3$  seeds.

### Pass criterion

$-k_B T \ln \langle e^{-\beta W} \rangle$  equals  $\Delta F$  within its bootstrap error at  $u = 0.005$  and  $0.01$ ; report the bias at  $u = 0.02$ .

## 8 Level 3 — geometry C: one gas drives a spring-loaded wall

### Why this step exists

First appearance of ordered energy. The observable of the whole paper,  $\epsilon = E_{\text{spring,max}}/W_{\text{in}}$ , is defined here in the simplest case where it has a prediction.

The free wall of mass  $M$  sits on a spring  $k$  and is pushed by the gas pressure. Linearise about equilibrium: the gas acts as a second spring of stiffness  $k_{\text{gas}} = \partial(PH)/\partial x = N_s k_B T [Z + \eta Z']/L_0^2$  (adiabatic:  $Z + \eta Z' + Z^2$  for fast motion), so the wall is an oscillator with

$$\omega_w^2 = \frac{k + k_{\text{gas}}}{M}, \quad T_w = \frac{2\pi}{\omega_w}. \quad (9)$$

A piston push of duration  $\tau_{\text{push}}$  raises the gas pressure by  $\Delta P$ . Two limits with closed forms:

$$\text{quasi-static } (\tau_{\text{push}} \gg T_w) : \quad x_{\text{eq}} = \frac{\Delta P H}{k + k_{\text{gas}}}, \quad E_{\text{spring}} = \frac{1}{2} k x_{\text{eq}}^2, \text{ no overshoot}; \quad (10)$$

$$\text{impulsive } (\tau_{\text{push}} \ll T_w) : \quad \Delta p_w = \int \Delta P H dt, \quad E_{\text{spring,max}} = \frac{k}{k + k_{\text{gas}}} \cdot \frac{\Delta p_w^2}{2M}. \quad (11)$$

In between,  $\epsilon(\tau_{\text{push}}/T_w)$  interpolates; that curve is the first result figure of Paper 2.

### 8.1 The apparatus is a PRE-LOADED spring, and that sets the ceiling

A spring that holds a wall against a gas is not at its natural length. It carries the standing gas force  $F = Nk_B T Z/L$ , so the wall sits  $d = F/k$  from the anchor and the stored energy is  $E = \frac{1}{2} k(d + s)^2$  for a further displacement  $s$ . Expanding,

$$\Delta E(s) = F s + \frac{1}{2} k s^2, \quad (12)$$

which is **first order** in  $s$ . The  $F s$  term is work done lifting a weight — the most classical definition of ordered work output there is, and exactly what a Szilard-type engine is judged by. The pre-load cancels out of the *dynamics*, because at  $s = 0$  the spring's pull balances the gas exactly, but it does not cancel out of the *energy*.

The two limits fix the shape of the answer:

- $k \rightarrow 0$  — a **pushrod**. The wall simply follows the piston,  $s \rightarrow \Delta x$  and  $\epsilon \rightarrow 1$  trivially. Nothing has been captured *from the gas*; the piston did the work directly.
- $k \rightarrow \infty$  — a **rigid wall**.  $s \rightarrow 0$  and  $\epsilon \rightarrow 0$ .

In between, the quasi-static capture is Eq. (12) evaluated at  $s_{\text{qs}} = \Delta x k_{\text{gas}}/(k + k_{\text{gas}})$ .

**The gas spring is adiabatic, and it is Paper 1's  $c_s$ .** There is no heat bath in this box, so a compression heats the gas and the restoring force rises faster than  $Z(\eta)$  alone says. With  $U = Nk_B T$  in 2D,  $dU = -P dV$  gives  $d \ln T / d \ln L = -Z$ , and

$$k_{\text{gas}}^{\text{ad}} = - \left. \frac{\partial F}{\partial L} \right|_S = \frac{Nk_B T}{L^2} (Z + \eta Z' + Z^2) = \frac{Nk_B T}{L^2} c_s^2, \quad (13)$$

*exactly Paper 1's measured sound speed.* Here that is 0.04938 against the isothermal 0.02457, a factor 2.01, and using the isothermal value put the model a factor 2.3 below the measured wall displacement. The adiabat cross-checks independently: it gives  $T_f/T_i = 1.1432$  for this 10 % compression against Level 1's measured 1.149, agreeing to 0.5 %.

**Note: why the 2026-09-19 pilot could not pass.** For an *unloaded* spring the  $F s$  term is absent and  $\Delta E = \frac{1}{2} k s^2$  is second order, maximal at  $k = k_{\text{gas}}$  with  $E_{\text{max}} = k_{\text{gas}} \Delta x^2 / 8 = Nk_B T f^2 c_s^2 / 8$  — 0.098  $k_B T$  at  $f = 0.1$ , against a  $k_B T / 2$  thermal pedestal. No choice of  $k$  rescues it. That is the pilot, and its numbers stand as the design finding. It also cannot be built: the static offset at  $k = k_{\text{gas}}$  is  $F/k_{\text{gas}} = L Z / (Z + \eta Z') = 0.816 L$  *independent of  $N$* , so the stiffness that maximises unloaded capture is the one at which the spring cannot be statically balanced inside its own apparatus.

**Units trap, fixed.** --spring-k is per **pixel**<sup>2</sup>, not per  $\sigma^2$  — a factor  $24^2 = 576$ . The pilot's  $k = 5$  was  $2880 k_B T / \sigma^2$ , so its quoted  $T_w = 39.5$  was really 1.66 and every cell it believed impulsive was quasi-static. Confirmed two ways: equipartition gives implied  $k = 2885$ , and the measured wall period is 1.657 against 1.656 predicted. Use --spring-k-sigma, which takes  $k_B T / \sigma^2$ ; --spring-k is kept unchanged so older commands still mean what they say.

**Verdict (2026-09-25).** With the bulk equation of state the parameter-free model predicts the settled wall displacement to  $+3.6 \pm 0.6\%$  (five cells pooled, measured  $\bar{s} = 0.8705 \pm 0.0049\sigma$ ). The residual is the finite-size stiffness of the 100-disk box, the same effect Paper 1 measures as  $+1\%$  in  $c_s$ , but it is *not independently pinned here*, so no attribution is claimed; the transient excess reported earlier was a peak statistic reproduced by a no-push control. **Level 3 closes on the observable; the quantitative close follows from the measured  $F(L)$  set**, which obtains  $Z_{\text{box}}(\eta) = FL/(Nk_B T)$  from the momentum delivered to a held wall and the gas temperature at the same instant, with no equation of state and no area convention assumed. The peak statistic is retained as a diagnostic only.

## 8.2 What replaces the impedance criterion

It is not impedance matching. The maximum of  $\epsilon_{\text{coh}}$  tracks  $\tau_{\text{push}}/T_w$ , not  $Z_w/Z_g$ : that is a driven oscillator, where a pressure rise applied faster than the wall's period overshoots up to twice the static displacement and one applied slower is followed quasi-statically. The acoustic pulse ( $Au^2 \approx 1 k_B T$  at  $u = 0.2$ ) is a small correction to a  $16 k_B T$  bulk pressure rise, so an impedance formula for the pulse was never going to be the controlling physics.

The prediction is a one-degree-of-freedom model with **no free parameter**:

$$M\ddot{s} = F_{\text{ad}}(L - x_p(t) + s) - F_{\text{ad}}(L) - ks, \quad x_p(t) = \min(ut, \Delta x), \quad s(0) = \dot{s}(0) = 0, \quad (14)$$

with  $F_{\text{ad}}$  the exact adiabatic gas force and  $E_{\text{max}} = \max_t \Delta E(s(t))$ .

### What we run (2026-09-21)

Master box, geometry C (divider removed, one gas of 100 disks,  $\eta = 0.1001$ ,  $\Delta x = 7.96\sigma = 10\%$ ), spring **pre-loaded** at wall  $+F/k = 33.65\sigma$ ,  $k = 0.5 k_B T/\sigma^2$ , step protocol, 40 seeds per cell, event log on, trace  $200\sigma$ -time past the stop.  $M_s \in \{2, 10, 50, 200, 1000\}$  at  $u \in \{0.2, 0.5, 1.0\}$ , plus  $u \in \{0.05, 0.1\}$  at  $M_s \in \{50, 200\}$  to anchor the quasi-static end: 760 runs.  $M_s$  is the discriminator because Eq. (12) contains no  $M_s$  while Eq. (14) does. Primary observable is the **ensemble-mean**  $E_{\text{spring}}(t)$ , baseline and peak-picking bias both taken from the causally blind window  $t < 0.4 L/c_s$  after the push starts.

### Pass criterion (2026-09-21)

(i) The ledger closes with the spring term to  $10^{-8}$ . (ii) The coherent peak stands  $> 3\sigma$  above the pedestal. (iii) Eq. (14) reproduces  $\Delta E_{\text{coh}}(M_s, u)$  within  $2\sigma$  at the speeds where the gas is genuinely quasi-static,  $\tau_{\text{push}} > L/c_s$  (i.e.  $u \leq 0.1$ ). At  $u \geq 0.2$  the deviation is *reported* as the acoustic contribution, measured rather than assumed, with no pass claimed.

## 9 Level 4 — geometry B: transmission through a free divider

### Why this step exists

Before stacking walls we need the transmission of one wall between two gases, and we already own half of this result: the slow mode of Paper 1 *is* the heat conduction of the divider.

Release the divider between gas 1 (pushed) and gas 2. Two channels carry energy across: the fast one, pressure waves moving the divider ( $\tau \sim \tau_{\text{sound}}$ ), and the slow one, heat conduction by divider collisions. Paper 1 measured the slow one: correlation time  $\tau_{\text{heat}} = 54\text{--}100$  acoustic periods at  $M = 100\text{--}1000$ , amplitude set by the isothermal stiffness. The adiabatic-piston literature predicts  $\tau_{\text{heat}} \propto M/m$ ; we can now test that with the  $KE_L, KE_R$  columns, and it becomes a figure for free.

The transmission observable is the energy that has crossed after the push has relaxed:

$$\mathcal{T}(t) = \frac{\Delta K E_{\text{gas}}(t)}{W_{\text{in}}}, \quad \mathcal{T}_{\text{fast}} = \mathcal{T}(t \approx 5\tau_{\text{sound}}), \quad \mathcal{T}_{\text{slow}} = \mathcal{T}(t \rightarrow \infty) \rightarrow \frac{1}{2} \text{ (equipartition of the excess heat)}. \quad (15)$$

The prediction to test for the fast channel is the two-body energy-transfer fraction with the gas column as a mass:

$$f(m_1, m_2) = \frac{4m_1 m_2}{(m_1 + m_2)^2}, \quad m_1 \rightarrow \text{effective gas mass } N_s m, \quad m_2 \rightarrow M, \quad (16)$$

which is heuristic and is exactly what Level 4 measures the validity of.

#### What we run

Geometry B,  $L_0 = 20$ ,  $N_s = 50$  each side,  $\eta = 0.196$ , held wall at  $L_0$  (mass  $10^9$ ), free divider at  $1.5L_0$  with  $M = 20, 50, 100, 200, 500, 1000, 2000$ ; piston step  $u/c_s = 0.1$ , travel  $2\sigma$ ; 25 seeds. Run to  $t = 200\tau_{\text{sound}}$  so  $\mathcal{T}_{\text{slow}}$  is reached.

#### Pass criterion

$\mathcal{T}_{\text{slow}} \rightarrow 0.5$  within  $2\sigma$  for every  $M$  (a ledger-level check).  $\tau_{\text{heat}}(M)$  from the exponential approach; report its exponent in  $M$  (expected +1).  $\mathcal{T}_{\text{fast}}(M)$ : compare with (16); the pass is not agreement but a *measured* curve with error bars and the location of its maximum in  $\alpha = M/N_s$ .

#### What we measured (2026-09-17)

**Level 4: the slow channel is already measured; the fast channel is not started.** From Paper 1, with the  $KE_L, KE_R$  columns: the slow divider mode is stationary (its variance is the same in both halves of a 1000-period record, ratio 0.72–1.25), survives a  $100\times$  longer hold, and has correlation time 53–102 acoustic periods at  $M = 100$ –1000. Its amplitude is the *isothermal* equipartition value  $\langle x^2 \rangle = L^2/(2N_s[Z + \eta Z'])$  to 4% for light dividers, while the adiabatic form overshoots by 37–62%. It follows  $T_L - T_R$  directly: correlation +0.978 ( $M = 100$ ) and +0.910 ( $M = 1000$ ) with the slow part of the compartment-temperature difference, positive sign (a hotter left compartment pushes the divider to  $+x$ ), lag  $\leq 0.25$  periods at  $M = 100$ , and the pressure balance  $x = (L_0/2)(\Delta T/T) Z/(Z + \eta Z')$  reproduces the amplitude to 6–7%. Across the A2 size ladder its amplitude falls as  $N^{-0.39}$  to  $N^{-0.54}$  in units of the box, i.e. it is a finite-size fluctuation that vanishes as  $N \rightarrow \infty$ . What is *not* yet measured:  $\tau_{\text{heat}}(M) \propto M$  over a mass ladder, and the fast channel  $\mathcal{T}_{\text{fast}}$  — both need the geometry-B runs of this level.

#### How to run it yourself

**Data:** Paper 1's A1 v2 campaign, already on disk — this level needs no new runs yet. **Analysis:** `python3 hspist3/validation/paper1.linewidth.20260918.py` gives  $\Gamma(\eta)$  from the divider linewidth; `damping.test.20260915.py` and `A2.slow.mode.20260915.py` give the slow-mode amplitude and its finite-size scaling.

## 10 Level 5 — geometry D and more walls: the efficiency experiment

#### Why this step exists

This is the paper's question: how much of the injected work arrives as ordered energy, and what the protocol, the masses and the number of walls do to it.

Observable, protocol-averaged over seeds:

$$\epsilon = \frac{E_{\text{spring,max}}}{W_{\text{in}}}, \quad 1 - \epsilon = \frac{\text{heat left in gases} + KE_{\text{walls at the peak}}}{W_{\text{in}}}. \quad (17)$$

Knobs and the prediction for each: piston speed  $u/c_s$  (shock heating above  $\approx 0.3$ ; Level 2 gives the friction), spring-wall mass and divider masses (impedance matching, Level 4), number of intermediate walls  $n_w$  (the collision-chain estimate: two masses  $m_1 \rightarrow m_2$  transfer  $f$  directly; with one intermediate  $\sqrt{m_1 m_2}$  the product of the two steps is larger, 0.36 vs 0.18 for  $50 \rightarrow 1000$ ; with geometric steps of ratio 2.7, 0.45). The prediction is therefore *ordered*: geometric intermediate masses raise  $\epsilon$ , arbitrary ones lower it.

#### What we run

Geometry D:  $L_0 = 20$ ,  $H = 10$ ,  $\eta = 0.196$  in each gas compartment, spring wall  $M_s = 1000$ ,  $k$  as in Level 3. Series 1: no intermediate wall,  $u/c_s = 0.02, 0.05, 0.1, 0.2, 0.5$ . Series 2: one divider,  $M_d = 50, 100, 224, 500, 1000$  at  $u/c_s = 0.1$ . Series 3:  $n_w = 0, 1, 2$  with geometric masses, and  $n_w = 2$  with two “wrong” masses (50, 50 and 1000, 1000), at  $u/c_s = 0.1$ . 25 seeds each. Every series carries the ledger.

#### Pass criterion

Series 1:  $\epsilon$  decreases monotonically with  $u/c_s$  and the loss  $1 - \epsilon$  at small  $u$  is consistent with  $\zeta$  from Level 2. Series 2:  $\epsilon(M_d)$  has a maximum; report its position against (16). Series 3:  $\epsilon(n_w = 2, \text{geometric}) > \epsilon(n_w = 1) > \epsilon(n_w = 0)$  and both wrong-mass chains below  $n_w = 0$ , each ordering at  $> 2\sigma$ . This ordering, with error bars, is the headline figure.

## 11 Level 6 — information (only after 0–5, and only with a stochastic protocol)

With a protocol  $x_t$  drawn from a Markov chain and a coarse-grained system state  $s_t$ , Still, Sivak, Bell and Crooks give

$$\beta \langle W_{\text{diss}}[x_t \rightarrow x_{t+1}] \rangle \geq I_{\text{mem}} - I_{\text{pred}} = I_{\text{nost}}, \quad (18)$$

left side from the ledger (Levels 0–2), right side from joint histograms over  $10^3$ – $10^4$  trajectories. Deterministic protocols have nothing to compute here, which is why Levels 0–5 use them and Level 6 cannot. This level is a separate paper’s worth; it is listed so the logging of Levels 0–5 is designed to serve it (per-interval work,  $x_t, s_t$ ).

## 12 Order, cost, and the two flags we need first

| Level         | Runs              | Needs                                      | Status                                                                      |
|---------------|-------------------|--------------------------------------------|-----------------------------------------------------------------------------|
| 0             | 0 (existing data) | nothing — the log exists                   | <b>passed:</b> $8 \times 10^{-12}$ (energy), $1 \times 10^{-13}$ (momentum) |
| 1             | 1430 done         | finite-box $Z$ along the path              | <b>passed:</b> $-0.015 \pm 0.006 k_B T$ , $1.7\sigma$                       |
| 2 (slow end)  | 760 done          | nothing new ( $\zeta$ from existing holds) | <b>passed, prediction corrected:</b> no linear term; $W - W_0$              |
| 2 (fast end)  | 300               | nothing new                                | not started                                                                 |
| 2 (Jarzynski) | $2 \times 10^3$   | canonical seeder flag                      | not started                                                                 |
| 3             | 125               | spring + $KE$ logging (exists)             | not started                                                                 |
| 4             | 175               | nothing new                                | slow-mode data already partly exists                                        |
| 5             | 375               | multi-wall driver                          | not started                                                                 |

Two code items, both logging or seeding, no physics: the per-event wall-impulse log (serves Level 0 momentum *and* Level 2  $\zeta$ ) and the canonical seeder (Jarzynski). Level 5 additionally needs the multi-wall driver. Everything else runs on the binary as it is. Total is about 3000 trajectories at  $L_0 \leq 40$ , i.e. hours, not days — the expensive part of Paper 2 is the analysis discipline, not the machine.

## The one sentence per level, for the paper

0. Energy and momentum are conserved to round-off in every accepted trajectory.
1. The slow-push limit of the work equals the quasi-static work of the finite box.
2. In a closed box the friction coefficient measured from equilibrium fluctuations integrates to zero at the sound-crossing time, so the excess work is second order in the piston speed and equals the kinetic energy of the compression flow; at high speed it is single-collision work with no free parameter.
3. A spring-loaded wall driven by one gas stores the predicted fraction of the work in its two limits.
4. A free divider transmits energy through a fast pressure channel and a slow heat channel, the latter being the mode found in Paper 1.
5. The fraction of work delivered as ordered energy depends on speed, mass and the number of walls in the predicted order.